MINI PROJECT PRESENTATION

Department of Computer Science and Engineering

Skillsphere College Student Interest (Domain) Based Clubs

(1st–4th Year)

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MARCH 2026

01 INTRODUCTION & CHALLENGES

Problem Statement

Many students are confused about which domain to choose in their college career.

There is no proper structured roadmap for learning technical skills.

Most college clubs conduct random events without continuous learning.

Junior students do not get proper guidance from senior students.

02 GOALS & ASPIRATIONS

Objectives

To provide structured domain selection for students

To implement a daily task-based learning system

To introduce a credit and leaderboard system for motivation

To enable 4th-year students to act as coordinators and mentors

To provide proper learning roadmaps for each domain

To improve collaboration between junior and senior students

Existing System

- Currently, students choose domains without proper guidance.
- There is no structured learning system available in most colleges.
- Clubs conduct events randomly without tracking skill development.
- There is no system to monitor student progress.

04 SKILLSPHERE PLATFORM

Proposed System

A web-based platform that provides structured domain-based learning for students from 1st to 4th year.

Students select domains based on their interests.

Senior students act as coordinators and mentors.

Daily tasks are assigned to improve consistency.

A credit and leaderboard system motivates students.

Student progress is tracked to ensure continuous skill development.

05 ROLE-BASED STRUCTURE

Architecture Workflow

1

Role Assignment

1st–3rd year students act as learners. 4th-year students act as coordinators/mentors.

2

Domain Selection

Students select their domain (Web Dev, AI/ML, Cybersecurity, Data Science, etc.)

3

Task Assignment

Coordinators assign daily tasks structured around the chosen domain.

4

Task Submission

Students submit task proof (GitHub link/files) for evaluation.

5

Reward & Track

Credits awarded upon approval. Leaderboard and progress tracking are updated automatically.

06 TECH STACK

Tools & Technologies

FRONTEND

HTML · CSS · JavaScript · React

BACKEND

Node.js · Express · Flask

DATABASE

MongoDB · MySQL

VERSION CONTROL

Git · GitHub

07 IMPLEMENTATION

Module Description

1

User Management

Student registration and login system

2

Domain Selection System

Students select domains based on interest

3

Daily Task System

Mentors assign tasks and students submit proof

4

Credit & Leaderboard System

Students earn credits and appear on the leaderboard

5

Progress Tracking System

Tracks student performance and consistency